High Performance 4nm FinFET Platform (4LPE) with Novel Advanced Transistor Level DTCO for Dual-CPP/HP-HD Standard Cells

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***Abstract*—**In this paper, a leading edge High performance 4nm FinFET Platform (4LPE) is demonstrated, featuring a novel advanced Transistor level DTCO (T-DTCO) in Middle-of-Line/Back-End-of-Line and careful process optimization, which enables dual-CPP/HP-HD standard cell with Gate contact-over-RX scheme. Compared to 5nm Platform (5LPE), Performance +7~10% and Power -12% gain are successfully obtained in both typical and low Vdd, as well as a flat layout dependence. 4LPE enhancement is accomplished by design-manufacturing favorable approaches. Moreover, newly adapted SRAM Ultra High-Density cell (UHD), Product IDDQ variation control and reliability verifications were completed for the technology delivery.

I. Introduction

4th Industrial revolution and the pandemic circumstance have been evolving Work/Life style change to create tremendous demands and opportunities for Semiconductor industry, such as HPC, PC and 5G Market [1]. The growth Forecast predicts a bright future of Semiconductor business. In contrast, it is also well-known that Processor Frequency enhancement has been retarded due to Power limitation and scaling difficulty. To maintain Moore's law, various innovative approaches have been implemented and proposed for the future nodes, such as Multi-Bridge-Channel (MBC) FET[2], CFET[3], and 2D-FET [4].

Besides the beyond FinFET, a continuous upgrade for the state-of-Art FinFET Platform has been highly demanded and critical to cover various product requirements. To enhance a product target, each application oriented optimization is indispensable. This requires dual-CPP/Track supply by Platform Technology to cover from High performance to Low power applications with careful process optimization for all cells co-centering [5], as well as the extended Platform in specific operation domain [6]. On the other hand, the extreme pitch scaling itself is definitely the key obstacle to boost performance by process optimization due to less stress sensitivity with smaller volume, more difficult short channel effect controllability and a tradeoff to manufacturability. Transistor level Design-Technology-Co-Optimization (T-DTCO), which is a part of Transistor design rather than Design optimization, is one of critical grips to overcome the scaling tradeoff without any cost-adder.

In this paper, 4nm FinFET Platform (4LPE) is revealed with a novel T-DTCO and other technology features as a leading edge Fin Technology.

II. Technology and Device Characteristics

A: Technology Definition and Trend

Despite of a hindrance of the scaling, all AC/DC/Power performance have been continuously improved by a new process/material/structure introduction as shown in Fig.1. On the other hand, a paradigm shift of the improvement methods (knobs) has been occurred with extremely scaled dimension. Fig.2 indicates a ratio of performance enhancement categories by (1) Conventional Process improvement: Fin structure, SD epi (stress, concentration), Materials, Thermal design etc., (2) Transistor Level DTCO (T-DTCO): Layout optimization to modulate Transistor level RC. As technology goes on, T-DTCO plays more significant role for the improvement to occupy ~50% portion of total boosting. Precise T-DTCO enables not only DC components (External Resistance (Rext), Tr channel Resistance (Rch)), but also AC components (Capacitance, AC_R_effective), which is more manufacturing favorable scheme. Another hurdle with scaled pitch is a loading effect. Although the loading effect reduction process was successfully delivered in 5nm [5], it is getting harder to satisfy co-optimization for both dual CPP/cell-height (HD/HP). T-DTCO certainly helps to reduce the burden for the process adjustment.

B: Performance Enhancement by T-DTCO and MOL Process

In Table I, the state-of-Art 4nm platform (4LPE) feature is shown and 4LPE uses the most advanced modules with full EUV process.

As described in previous section, T-DTCO is one of key items to enhance the performance to avoid yield tradeoff (Manufacturing friendly). Fig.3 shows T-DTCO Frequency enhancement result for 4LPE HD/HP cell by Middle-of-Line (MOL) and Back-End-of-Line (BEOL, M1 level). MOL T-

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DTCO effectively adjusts contact R/C optimization combined with FEOL process interaction to obtain up to 3% boosting, including AC_R_effective. Together with BEOL T-DTCO (BEOL design itself, MOL-M1 level interface optimization), HD/HP cell performance are successfully enhanced by totally ~3/5%, respectively. In addition to T-DTCO, newly introduced Gate Contact-Over-RX design rule for area scaling requires careful MOL process setting with the tight gate pitch. Through Contact profile (slope/recess) with suitable etch adjustment, a few % AC gain has been achieved by Capacitance reduction without the drivability penalty (Fig.4).

Overall Performance/Power gain are shown in Fig.5 for both typical and low Vdd range. Compared to 5nm Platform [5], Significant gain for Performance +7~+10% and Power -12% are accomplished in typical ~ low Vdd range through Manufacturing favorable knobs.

C: Layout Effect

Besides PPA enhancement, another essential item is Layout Effect and a flatter layout dependence is strongly favorable to provide design friendly device for total Chip performance improvement. A performance enhancement by stress effect possibly causes a strong layout effect without careful process setting [7].

Fig.6(a) shows Layout schematic of 2 crucial items for 4LPE, such as Diffusion Break - DUT distance (SAB) and Gate contact - RX boundary distance. Thanks to careful design of Stress process and the utilization of T-DTCO, almost flat SAB dependence is realized (Fig.6(b)) and Transistor performance itself is well matched between minimum SAB Tr and Standard-cell like SAB Tr, which indicates no stress relaxation toward Single-DB (Fig.6(c)). Simultaneously, by preventing etching gas penetration to Work-Function metal, the insensitive contact position dependence (< ±10mV shift) is achieved for different Vt flavors with various WF metal condition, no matter of the gate contact location "On" or "Off" Transistor (Fig.6(d)).

T-DTCO enhanced devices with the flat layout dependence allow 4LPE a benefit of a reduction of manufacturing and design burden.

III. Other 4LPE Baseline Characteristics

Besides Transistor characteristics, other 4LPE Platform technology features are describes in this section.

A: SRAM

Newly Ultra-High-Density SRAM cell has been introduced in 4LPE. Butterfly curves and Vmin distribution comparisons are shown in Fig.7 and 8 for UHD and HD cells. Even with scaled cell size, UHD has no penalty for SNM window, Vmin, and a variation, thanks to the optimized process to minimize layout difference.

B: Product IDDQ

For a manufacturing readiness, Chip level IDDQ control is indispensable and governed by a superior short-channel-effect. Compared to 5LPE, 4LPE SCE is deliberately managed by a structure modification to achieve DIBL and Off-current (Isoff) at fixed Vth (~Swing) by 10/20% reduction, especially in PFET (Fig.9). As a result of the good SCE, Product level IDDQ has been tightened by 0.76x compared to the early stage, which is already 0.97x better than 5LPE mass production level (Fig.10).

C: Reliability

Finally, 4nm reliability assessment is stated in Table II. Thanks to taking an advantage of T-DTCO, 4LPE process imposes less burden on the reliability even with the drivability enhancement. In conclusion, most of reliability items are improved and all reliability criteria have been passed with enough margin, compared to 5LPE.

IV. Conclusion

In this paper, the paradigm shift of Device level performance enhancement method is revealed first time for the extremely scaled nodes. Following the trend, the production - Design friendly advanced 4LPE Platform technology is presented for High performance/Low power applications. The technology introduces the enhanced T-DTCO to modulate both Resistance/Stress effect and Capacitance and successfully provides RO Performance +7~+10% and Power -12% gain without any cost/yield conflict. 4LPE technology is ready for the mass-production with other superior Layout effect, SRAM stability, IDDQ variation and reliability behaviors.

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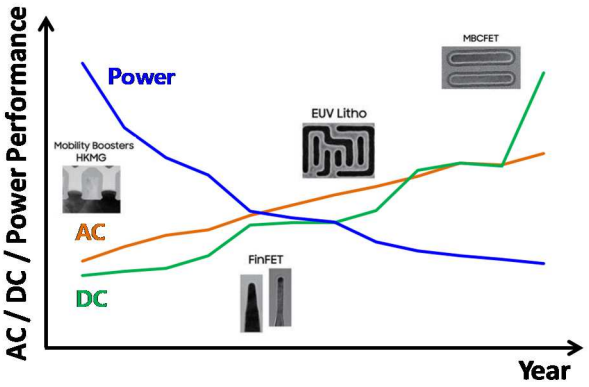


Fig.1: AC/DC /Power Performance Technology trend

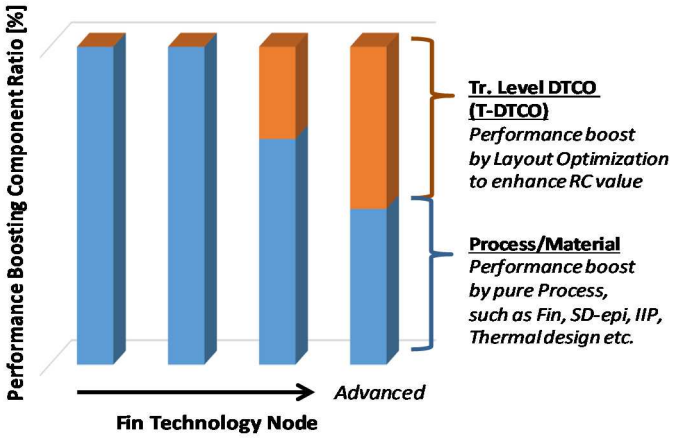


Fig.2: Comparison of Performance boosting component ratio by Transistor Level DTCO (T-DTCO) v.s. Process/Material in each FinFET technology.

Table I: Key process feature for 4nm technology with the most advanced Fin/SD module and Full EUV process

Key Feature	8nm	7nm (2nd gen.)	5nm (5LPE)	4nm (4LPE)
Fin	4th gen.	5th gen.	6th gen.	7th gen.
S/D	4th gen.	5th gen.	6th gen.	7th gen.
Contact /BEOL	MPT	EUV	EUV	EUV
Process Future	MDB/SDB Dual CPP	MDB/SDB Dual CPP	MDB/SDB Dual CPP Contact -on-RX edge	SDB Dual CPP Contact -over-RX
Reference	VLSI 2018'	VLSI 2018'	IEDM 2020'	This work

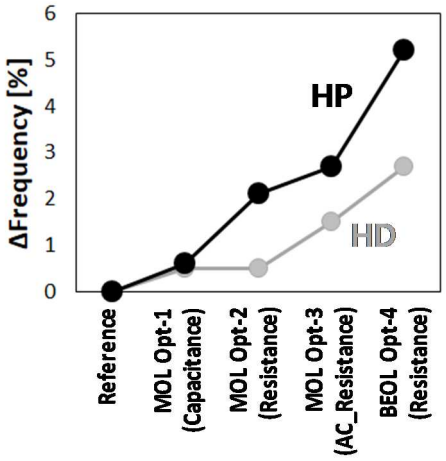


Fig.3: Performance improvement by T-DTCO (MOL/BEOL Optimization) for both HD and HP cell by T-DTCO.

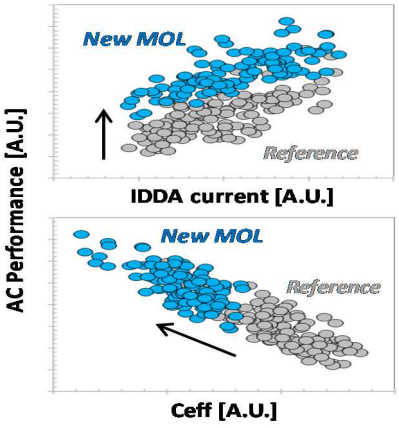
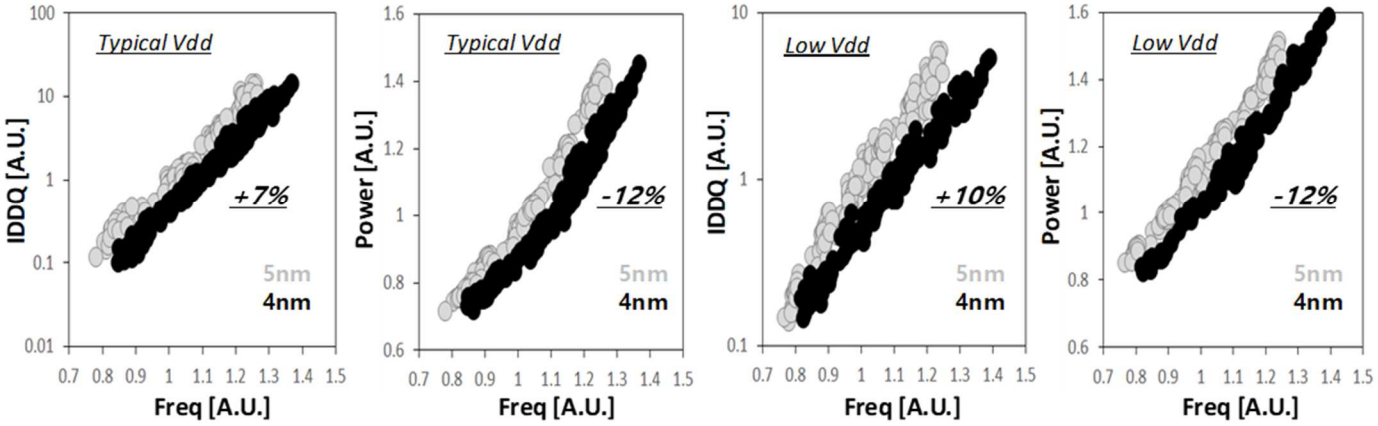


Fig.4: AC performance comparison by IDDA/Ceff between Original v.s. New Middle-of-Line (MOL) Process with Gate Contact-over-RX scheme.

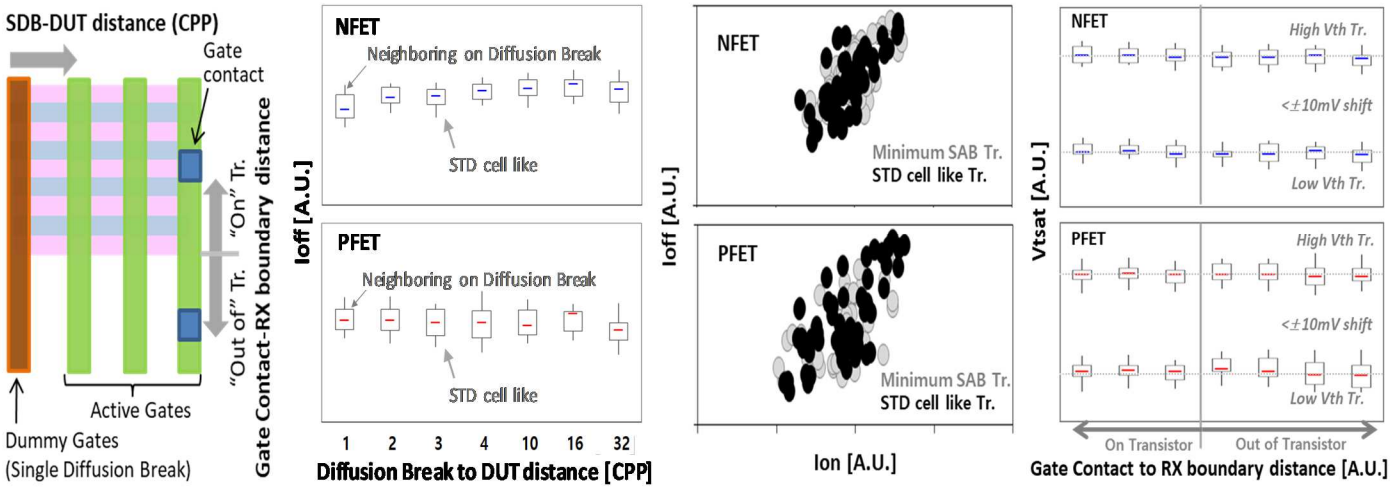


(a) Frequency-IDDA/Power comparison at Typical Vdd

(b) Frequency-IDDA/Power comparison at Low Vdd

Fig. 5: Logic Transistor Performance/Power comparison between 5nm (5LPE) [5] and 4nm (4LPE) Platform Technology with High density Cell at Typical/Low-Vdd.

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(a) Layout schematic for Diffusion Break-DUT (SAB) and Contact on gate distance

(b) SAB dependence of Ioff current for N/PFET from 1CPP (Min. SAB) to 32CPP

(c) DC comparison between minimum SAB and STD cell like Tr

(d) Contact on Gate position dependence of N/PFET Vth with different Tr flavor

Fig. 6: 4LPE Layout dependence for Diffusion Break to DUT distance (SAB dependence) and Contact on PC to DUT distance

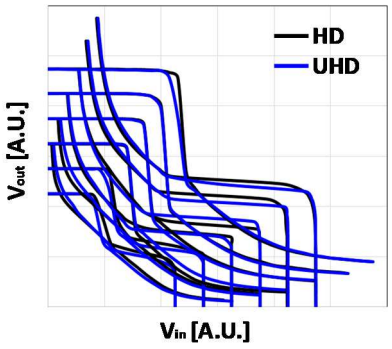


Fig.7: SRAM Butterfly curves for High Density (HD) and Ultra High Density (UHD) cell

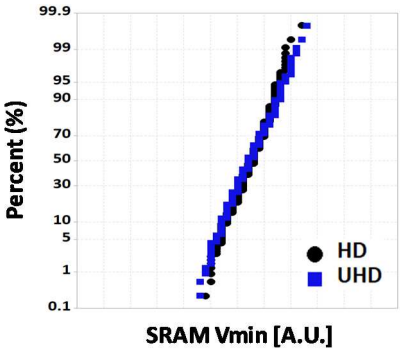


Fig.8: SRAM Vmin distribution for High Density (HD) and Ultra High Density (UHD) cell

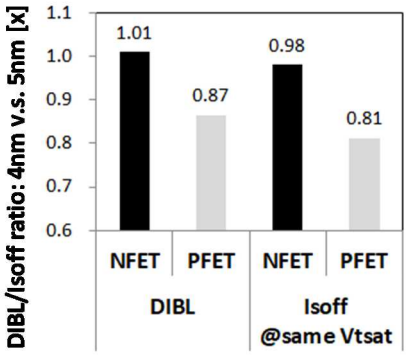


Fig.9: DIBL and Isoff ratio comparison between 5nm v.s. 4nm for High-Density cell

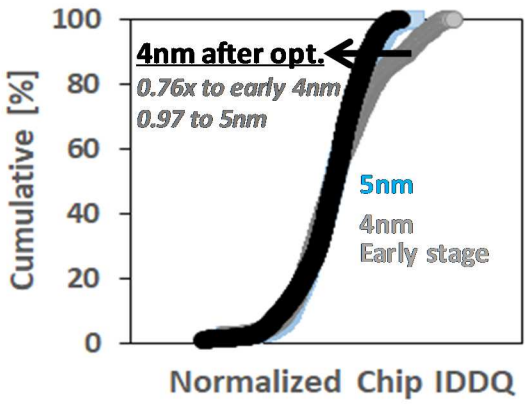


Fig. 10: Chip level IDDQ distribution improvement from 4LPE early stage to Optimized Process (Current)

Table II: Reliability statement for both thin (Core) and thick oxide (IO) transistor, compared to 5nm

Reliability Item			5nm → 4nm	
HCI	Thin Oxide	NFET	0.94x	Pass
		PFET	0.79x	Pass
	Thick Oxide	NFET	1.03x	Pass
		PFET	0.87x	Pass
BTI	Thin Oxide	NFET	1.15x	Pass
		PFET	0.72x	Pass
	Thick Oxide	NFET	0.85x	Pass
		PFET	0.94x	Pass

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